Project Part 2

1.3.1 Functional Requirements

FR1: The system shall allow secure account creation and login (email/password or SSO).

FR2: The system shall present a Placement Test to determine the initial CEFR proficiency level.

FR3: The system shall assign a CEFR level (A1-C1) based on the Placement Test results.

FR4: The system shall assign and display trackable daily goals (Today's Goal).

FR5: The system shall present a 10-question core lesson (Daily Lesson) with diverse question types (MCQ, matching, gap-fill, picture-to-word).

FR6: The system shall offer a 20-second time-limited 10-question Grammar Sprint.

FR7: The system shall provide a 30-second mini-game for vocabulary matching and contextual use (Word Sprint+).

FR8: The system shall display the score, time, and correct/incorrect answers upon lesson completion.

FR9: The system shall automatically increase the student's XP score upon successful completion of a task.

FR10: The system shall automatically unlock a Lesson Pack upon meeting the level prerequisites.

FR11: The system shall enable a 5-turn voice role-play session (Prepare Me) based on a selected scenario (e.g., Job Interview, Restaurant).

FR12: The system shall provide instant, basic feedback (mini scores) on the voice response in Prepare Me.

FR13: The system shall present a 5-question free-form conversation session (Talk Loop).

FR14: The system shall provide detailed scores (Fluency, Grammar, and Vocab) with a 3-point explanation after Talk Loop.

FR15: The system shall record and analyze fluency metrics (response time, pauses) during the Talk Loop session.

FR16: The system shall generate a weekly "focus list" of 10-20 words based on the student's recent mistakes (AI Vocabulary Advisor).

FR17: The system shall automatically compile a 10-question repeated practice session (Review Mode) based on recent mistakes and forgotten words.

FR18: The system shall send personalized reminders (e.g., "Don't break your streak!") based on activity patterns (Smart Reminder).

FR19: The system shall display skill-based scores, the level bar, and progress charts in the My Progress section.

FR20: The system shall allow the student to download the Weekly Progress Report as a PDF from the My Progress screen.

FR21: The system shall award automatic badges (Achievements) based on milestones like "7-Day Streak".

FR22: The system shall assign and track progress for weekly tasks (Weekly Challenge).

FR23: The system shall provide a Study Planner for selecting weekly study days and topics.

FR24: The system shall check all level-up conditions.

FR25: The system shall present a Transition Test to confirm level advancement.

FR26: The system shall allow students to track their Level Up progress with a visual bar.

FR27: The system shall allow users to update their password and email via the Settings screen.

FR28: The system shall provide a microphone test/setup tool in the Settings screen.

FR29: The system shall display the number of completed goals and adherence rate in the Study Planner.

FR30: The system shall integrate AI Vocabulary Advisor words directly into the Word Sprint+ flow.

FR31: The system shall show weak grammar topic tags (e.g., Tense, Preposition) upon Grammar Sprint completion.

FR32: The system shall support all question types listed in the Core Modules (MCQ, matching, gap-fill, picture-to-word).

FR33: The system shall categorize content into Lesson Packs based on CEFR levels (A1, A2, B1).

FR34: The system shall automatically award a "Review Badge" upon completion of Review Mode.

FR35: The system shall send behavior-triggered push/email notifications based on the Smart Reminder rules.

FR36: The system shall show "better say" suggested sentences after a Prepare Me response.

FR37: The system shall provide a dashboard for the System Administrator to monitor aggregate user statistics.

FR38: The system shall allow the System Administrator to update lesson content and metadata via a dedicated interface.

FR39: The system shall allow students to add new vocabulary acquired during Word Sprint+ to their personal word list.

FR40: The system shall display a "Congratulations" screen and unlock new content upon successful level advancement.

1.3.2 Non-Functional Requirements

NFR1: The user interface shall be highly intuitive and consistent across all modules so that a first-time user can start any core lesson (e.g., Daily Lesson or Grammar Sprint) within a maximum of 3 clicks from the main dashboard without guidance.

NFR2: The system shall process and display the scoring and correct/incorrect feedback for core lesson submissions (e.g., Daily Lesson, Grammar Sprint) within a maximum of 3second.

NFR3: The system shall store user passwords using a secure hashing algorithm and ensure all transmission of sensitive data (such as passwords and emails) is done over an encrypted protocol (HTTPS).

NFR4: system shall not lose data from the user's last completed lesson or test attempt in the event of an application crash or short-term outage.

NFR5: The system shall ensure that interface elements (buttons, text, images) are displayed correctly and legibly across different screen sizes (mobile and desktop).

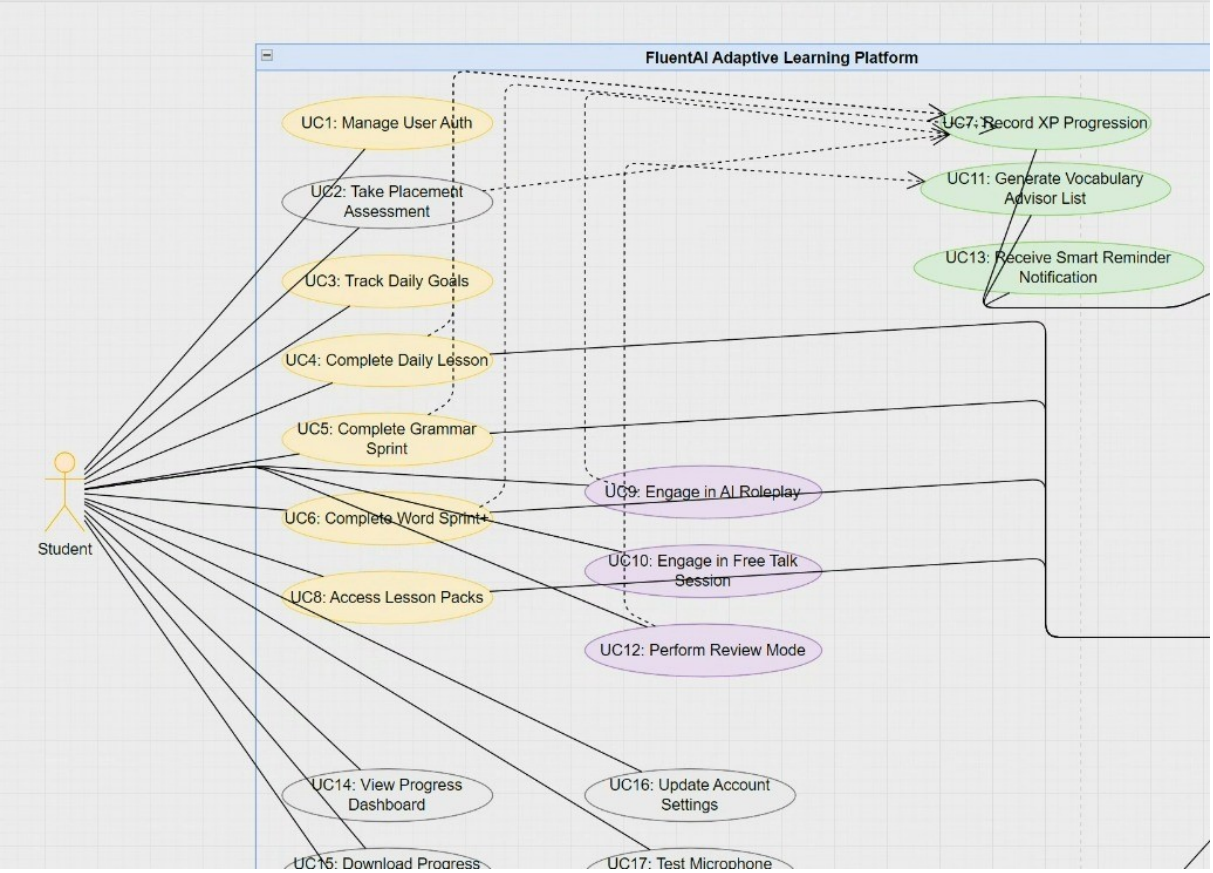
2. Define and Describe Use Cases

2.1 Define User-Goal Level Use Cases

Use Case ID	Use Case Title	Corresponds to FR	Primary Actor
UC1	Manage User Authentication	FR1	Student
UC2	Take Placement Assessment	FR2, FR3	Student
UC3	Track Daily Goals	FR4	Student
UC4	Complete Daily Lesson	FR5, FR8	Student
UC5	Complete Grammar Sprint	FR6	Student
UC6	Complete Word Sprint+	FR7	Student
UC7	Record XP Progression	FR9	AI Adaptive Engine
UC8	Access Lesson Packs	FR10	Student
UC9	Engage in AI Roleplay	FR11, FR12	Student

Use Case ID	Use Case Title	Corresponds to FR	Primary Actor
UC10	Engage in Free Talk Session	FR13, FR14, FR15	Student
UC11	Generate Vocabulary Advisor List	FR16	AI Adaptive Engine
UC12	Perform Review Mode	FR17	Student
UC13	Receive Smart Reminder Notification	FR18	AI Adaptive Engine
UC14	View Progress Dashboard	FR19	Student
UC15	Download Progress Report	FR20	Student
UC16	Update Account Settings	FR27	Student
UC17	Test Microphone Functionality	FR28	Student
UC18	Manage Study Planner	FR23	Student
UC19	Monitor Administrator Statistics	FR37	System Administrator
UC20	Manage Content Repository	FR38	System Administrator

2.2 Draw an UML Use Case Diagram



2.3 Write Use Cases in Brief Format

UC1: Manage User Authentication

- Title: Manage User Authentication
- Primary Actor: Student
- Goal: To gain secure access to the platform.
- Brief Description: The student logs in or registers securely, and the system verifies access rights.

UC2: Take Placement Assessment

- Title: Take Placement Assessment
- Primary Actor: Student
- Goal: To determine the initial CEFR level.
- Brief Description: The student completes the diagnostic test, and the system assigns the starting proficiency level.

UC3: Track Daily Goals

- Title: Track Daily Goals
- Primary Actor: Student
- Goal: To monitor progress against daily learning targets.
- Brief Description: The student views assigned daily tasks and sees completion counters update.

UC4: Complete Daily Lesson

- Title: Complete Daily Lesson
- Primary Actor: Student
- Goal: To master core concepts through structured questions.
- Brief Description: The student completes the 10-question lesson and receives a score and detailed feedback.

UC5: Complete Grammar Sprint

- Title: Complete Grammar Sprint
- Primary Actor: Student
- Goal: To practice grammar quickly under time pressure.
- Brief Description: The student attempts 10 timed grammar questions and receives a session score.

UC6: Complete Word Sprint+

- Title: Complete Word Sprint+
- Primary Actor: Student
- Goal: To rapidly acquire and reinforce vocabulary.
- Brief Description: The student plays the timed word game and adds newly learned words to their list.

UC7: Record XP Progression

- Title: Record XP Progression
- Primary Actor: AI Adaptive Engine
- Goal: To reward activity and track engagement.
- Brief Description: The system automatically calculates and saves earned XP after any completed activity.

UC8: Access Lesson Packs

- Title: Access Lesson Packs
- Primary Actor: Student
- Goal: To access structured content tailored to their level.
- Brief Description: The student browses available content packs, and the system unlocks relevant packs based on their level.

UC9: Engage in AI Roleplay

- Title: Engage in AI Roleplay
- Primary Actor: Student
- Goal: To practice speaking in a specific real-world scenario.
- Brief Description: The student provides voice responses to AI prompts and receives instant mini-scores for their speaking.

UC10: Engage in Free Talk Session

- Title: Engage in Free Talk Session
- Primary Actor: Student
- Goal: To develop fluency and receive detailed multi-metric analysis.
- Brief Description: The student responds to 5 open-ended questions and receives comprehensive Fluency, Grammar, and Vocab scores.

UC11: Generate Vocabulary Advisor List

- Title: Generate Vocabulary Advisor List
- Primary Actor: AI Adaptive Engine
- Goal: To provide personalized vocabulary remediation.
- Brief Description: The system analyzes past mistakes and automatically generates a focus list of difficult words.

UC12: Perform Review Mode

- Title: Perform Review Mode
- Primary Actor: Student
- Goal: To reinforce concepts missed in previous sessions.
- Brief Description: The student completes a 10-question quiz automatically compiled from their recent mistakes.

UC13: Receive Smart Reminder Notification

- Title: Receive Smart Reminder Notification
- Primary Actor: AI Adaptive Engine
- Goal: To motivate and guide the student based on learning patterns.
- Brief Description: The system sends personalized push notifications based on predefined behavioral rules (e.g., streak alerts).

UC14: View Progress Dashboard

- Title: View Progress Dashboard
- Primary Actor: Student
- Goal: To monitor overall progress and skill scores.
- Brief Description: The student views the My Progress screen showing skill scores, the level bar, and performance charts.

UC15: Download Progress Report

- Title: Download Progress Report
- Primary Actor: Student
- Goal: To obtain a formal summary of weekly performance.
- Brief Description: The student requests and downloads their weekly achievement summary as a PDF document.

UC16: Update Account Settings

- Title: Update Account Settings
- Primary Actor: Student
- Goal: To modify personal preferences or credentials.
- Brief Description: The student accesses the Settings menu to change their password, email, or other preferences.

UC17: Test Microphone Functionality

- Title: Test Microphone Functionality
- Primary Actor: Student
- Goal: To verify speaking module readiness.
- Brief Description: The student runs a quick test in the settings to ensure the microphone is working and accessible for voice input.

UC18: Manage Study Planner

- Title: Manage Study Planner
- Primary Actor: Student
- Goal: To structure their weekly learning schedule.
- Brief Description: The student defines study days and topics for the week and tracks completion status.

UC19: Monitor Administrator Statistics

- Title: Monitor Administrator Statistics
- Primary Actor: System Administrator
- Goal: To track aggregate system usage and health.
- Brief Description: The Administrator views a dedicated dashboard showing system-wide metrics and user statistics.

UC20: Manage Content Repository

- Title: Manage Content Repository
- Primary Actor: System Administrator
- Goal: To maintain and update all learning content.
- Brief Description: The Administrator uses the interface to add, edit, or remove lesson content, word lists, and scenario data.

2.4 Fully Dressed Use Cases

UC1: Manage User Authentication
Title: Manage User Authentication

Primary Actor: Student

Goal in Context: Ensures users can securely access their accounts and the platform.

Stakeholders and Interests:

- • Student: Wants secure access to personal progress and data.

Preconditions: System is accessible, and the user is on the login/register screen.

Success Guarantee: User successfully logs in or registers, gaining access to the main dashboard.

Main Success Scenario:

- ◦ Student enters credentials and clicks login/register.
- ◦ System verifies credentials.
- ◦ If valid, system grants access and displays dashboard.

Extensions:

- ◦ 2a. Invalid credentials → System displays error message.
- ◦ 3a. Network error → Retry prompt appears.

UC2: Take Placement Assessment
Title: Take Placement Assessment

Primary Actor: Student

Goal in Context: Determines the student's initial CEFR proficiency level.

Stakeholders and Interests:

- • Student: Wants accurate initial placement.
- • System: Needs correct data to assign levels.

Preconditions: User is logged in and has not yet taken the placement test.

Success Guarantee: Student's CEFR level is determined and saved.

Main Success Scenario:

- ◦ Student starts placement test.
- ◦ System presents test questions.
- ◦ Student completes and submits test.
- ◦ System analyzes results and assigns CEFR level.

Extensions:

- ◦ 3a. Student exits early → Progress saved.
- ◦ 4a. Server error → Results queued.

UC3: Track Daily Goals

Title: Track Daily Goals

Primary Actor: Student

Goal in Context: Allows students to monitor progress toward their daily learning targets.

Stakeholders and Interests:

- • Student: Wants to see daily goals and progress.

Preconditions: User is logged in and has completed at least one task.

Success Guarantee: Daily progress data is updated and displayed.

Main Success Scenario:

- ◦ Student opens Daily Goals section.
- ◦ System fetches goal data.
- ◦ System updates completion counters.

Extensions:

- ◦ 2a. No internet → Cached data shown.
- ◦ 3a. Data sync failure → Retry option displayed.

UC4: Complete Daily Lesson

Title: Complete Daily Lesson

Primary Actor: Student

Goal in Context: Enables mastery of core concepts through interactive lessons.

Stakeholders and Interests:

- • Student: Wants to complete daily learning tasks.

Preconditions: User has access to a lesson pack.

Success Guarantee: Lesson completed and feedback shown.

Main Success Scenario:

- ◦ Student opens Daily Lesson.
- ◦ System displays 10 questions.
- ◦ Student answers all questions.
- ◦ System provides feedback and score.

Extensions:

- ◦ 3a. Timeout → Lesson auto-submitted.
- ◦ 4a. Incomplete answers → Prompt for completion.

UC5: Complete Grammar Sprint

Title: Complete Grammar Sprint

Primary Actor: Student

Goal in Context: Helps students practice grammar quickly under time pressure.

Stakeholders and Interests:

- • Student: Wants rapid grammar practice.

Preconditions: User is logged in and has access to the grammar module.

Success Guarantee: Grammar sprint results saved and feedback shown.

Main Success Scenario:

- ◦ Student starts Grammar Sprint.
- ◦ System displays timed questions.
- ◦ Student answers all questions within time limit.
- ◦ System records results and displays feedback.

Extensions:

- ◦ 2a. Timer failure → Retry session.
- ◦ 3a. User exits early → Progress lost.

UC6: Complete Word Sprint+

Title: Complete Word Sprint+

Primary Actor: Student

Goal in Context: Provides a gamified vocabulary learning experience.

Stakeholders and Interests:

- • Student: Wants fun and effective vocabulary practice.

Preconditions: User has unlocked Word Sprint+ mode.

Success Guarantee: Session completed and results displayed.

Main Success Scenario:

- ◦ Student starts Word Sprint+.
- ◦ System shows timed vocabulary matches.
- ◦ Student completes all rounds.
- ◦ System displays score and allows word saving.

Extensions:

- ◦ 2a. Timeout → Partial score saved.
- ◦ 4a. Server error → Session results not uploaded.

UC7: Record XP Progression

Title: Record XP Progression

Primary Actor: AI Adaptive Engine

Goal in Context: Tracks engagement and progress by awarding XP.

Stakeholders and Interests:

- • AI Engine: Ensures user progress is accurately tracked.

Preconditions: User completes an activity.

Success Guarantee: XP successfully updated and recorded.

Main Success Scenario:

- ◦ System detects activity completion.
- ◦ XP value calculated.
- ◦ XP added to user profile.

Extensions:

- ◦ 2a. Connection lost → Update queued.
- ◦ 3a. Data conflict → Most recent entry kept.

UC8: Access Lesson Packs

Title: Access Lesson Packs

Primary Actor: Student

Goal in Context: Allows students to view and unlock structured content packs.

Stakeholders and Interests:

- • Student: Wants level-based lesson packs.

Preconditions: User logged in and at a defined CEFR level.

Success Guarantee: Lesson packs displayed based on student's level.

Main Success Scenario:

- ◦ Student navigates to Lesson Packs.
- ◦ System checks eligibility and unlocks packs.
- ◦ User opens selected pack.

Extensions:

- ◦ 2a. Pack locked → Display requirements.
- ◦ 3a. Load error → Retry suggested.

UC9: Engage in AI Roleplay

Title: Engage in AI Roleplay

Primary Actor: Student

Goal in Context: Allows speaking practice in real-world scenarios with AI feedback.

Stakeholders and Interests:

- • Student: Wants to improve speaking skills.

Preconditions: Microphone functional and permissions granted.

Success Guarantee: AI roleplay completed and feedback displayed.

Main Success Scenario:

- ◦ Student selects scenario and starts roleplay.
- ◦ System prompts with questions.
- ◦ Student responds verbally.
- ◦ AI analyzes and gives feedback.

Extensions:

- ◦ 2a. Mic not working → Error message.
- ◦ 4a. Speech not recognized → Retry prompt.

UC10: Engage in Free Talk Session

Title: Engage in Free Talk Session

Primary Actor: Student

Goal in Context: Provides open conversation practice with AI scoring.

Stakeholders and Interests:

- • Student: Wants deeper speaking assessment.

Preconditions: User logged in and has microphone setup.

Success Guarantee: Session completed and detailed scores displayed.

Main Success Scenario:

- ◦ Student starts Free Talk session.
- ◦ System asks open-ended questions.
- ◦ Student responds to each.
- ◦ System evaluates fluency, grammar, and vocab.

Extensions:

- ◦ 2a. Mic failure → Prompt to test.
- ◦ 3a. AI error → Session restarted.

UC11: Generate Vocabulary Advisor List

Title: Generate Vocabulary Advisor List

Primary Actor: AI Adaptive Engine

Goal in Context: Creates personalized vocabulary list based on user mistakes.

Stakeholders and Interests:

- • AI Engine: Improves personalized learning experience.

Preconditions: User has completed multiple lessons.

Success Guarantee: New focus list generated.

Main Success Scenario:

- ◦ System analyzes recent errors.
- ◦ System compiles difficult words.
- ◦ List displayed to user.

Extensions:

- ◦ 1a. No mistakes found → Empty list.
- ◦ 3a. Sync issue → Delayed update.

UC12: Perform Review Mode

Title: Perform Review Mode

Primary Actor: Student

Goal in Context: Reinforces previous mistakes through targeted review questions.

Stakeholders and Interests:

- • Student: Wants focused revision practice.

Preconditions: User completed lessons available for review.

Success Guarantee: Review session completed and recorded.

Main Success Scenario:

- ◦ Student opens Review Mode.
- ◦ System loads past errors as questions.
- ◦ Student answers them.
- ◦ System provides results.

Extensions:

- ◦ 2a. Incomplete data → Fallback questions shown.
- ◦ 4a. Submission error → Retry option.

UC13: Receive Smart Reminder Notification

Title: Receive Smart Reminder Notification

Primary Actor: AI Adaptive Engine

Goal in Context: Motivates user activity through personalized reminders.

Stakeholders and Interests:

- • AI Engine: Encourages consistent learning.

Preconditions: User has registered notification permissions.

Success Guarantee: User receives tailored notifications.

Main Success Scenario:

- ◦ System analyzes inactivity data.
- ◦ AI triggers suitable reminder.
- ◦ Notification delivered to user.

Extensions:

- ◦ 2a. Notification blocked → Log warning.
- ◦ 3a. Delay in delivery → Sent later.

UC14: View Progress Dashboard

Title: View Progress Dashboard

Primary Actor: Student

Goal in Context: Displays all metrics related to user's performance and progress.

Stakeholders and Interests:

- • Student: Wants to see visual progress data.

Preconditions: User logged in with recorded progress.

Success Guarantee: Dashboard loads correctly with visual analytics.

Main Success Scenario:

- ◦ Student opens My Progress screen.
- ◦ System retrieves and displays scores, bars, charts.

Extensions:

- ◦ 2a. No data available → Empty state shown.
- ◦ 3a. Server delay → Loading spinner displayed.

UC15: Download Progress Report

Title: Download Progress Report

Primary Actor: Student

Goal in Context: Allows users to export weekly progress reports as PDFs.

Stakeholders and Interests:

- • Student: Wants downloadable proof of progress.

Preconditions: User has at least one completed week.

Success Guarantee: Report downloaded successfully.

Main Success Scenario:

- ◦ Student requests report.

- ◦ System generates PDF.
- ◦ File downloaded to device.

Extensions:

- ◦ 2a. PDF generator fails → Retry prompt.
- ◦ 3a. No data available → Alert shown.

UC16: Update Account Settings

Title: Update Account Settings

Primary Actor: Student

Goal in Context: Allows users to modify credentials and preferences.

Stakeholders and Interests:

- • Student: Wants to change account details securely.

Preconditions: User logged in and on Settings screen.

Success Guarantee: Account successfully updated.

Main Success Scenario:

- ◦ Student edits settings fields.
- ◦ System validates inputs.
- ◦ System saves changes.

Extensions:

- ◦ 2a. Invalid input → Warning message.
- ◦ 3a. Save error → Retry request.

UC17: Test Microphone Functionality

Title: Test Microphone Functionality

Primary Actor: Student

Goal in Context: Ensures microphone works properly before speaking modules.

Stakeholders and Interests:

- • Student: Wants to confirm voice setup.

Preconditions: Device has microphone connected.

Success Guarantee: Test completed and status displayed.

Main Success Scenario:

- ◦ Student runs mic test.
- ◦ System records short sample.
- ◦ System confirms functionality.

Extensions:

- ◦ 2a. No input detected → Retry suggested.
- ◦ 3a. Permission denied → Request access.

UC18: Manage Study Planner

Title: Manage Study Planner

Primary Actor: Student

Goal in Context: Allows creation and tracking of weekly study schedules.

Stakeholders and Interests:

- • Student: Wants organized learning routines.

Preconditions: User logged in and plan editable.

Success Guarantee: Planner updated and displayed.

Main Success Scenario:

- ◦ Student opens Study Planner.
- ◦ Student selects study days and topics.
- ◦ System saves and displays schedule.

Extensions:

- ◦ 2a. Invalid date → Prompt correction.
- ◦ 3a. Save failure → Retry action.

UC19: Monitor Administrator Statistics

Title: Monitor Administrator Statistics

Primary Actor: System Administrator

Goal in Context: Enables monitoring of system health and aggregate user data.

Stakeholders and Interests:

- • Admin: Wants real-time usage data.

Preconditions: Admin logged in to dashboard.

Success Guarantee: Statistics displayed successfully.

Main Success Scenario:

- ◦ Admin opens dashboard.
- ◦ System retrieves usage and error logs.
- ◦ System visualizes metrics.

Extensions:

- ◦ 2a. Metric unavailable → Display placeholder.
- ◦ 3a. Timeout → Refresh option.

UC20: Manage Content Repository

Title: Manage Content Repository

Primary Actor: System Administrator

Goal in Context: Allows administrators to add, edit, or delete content items.

Stakeholders and Interests:

- • Admin: Wants full control of lesson content.

Preconditions: Admin logged in with permissions.

Success Guarantee: Content updated successfully.

Main Success Scenario:

- ◦ Admin selects Manage Content.
- ◦ Admin performs add/edit/delete actions.
- ◦ System updates repository and confirms changes.

Extensions:

- ◦ 2a. Invalid input → Error prompt.
- ◦ 3a. Save conflict → Retry option.